



RAK11310 WisBlock LoRaWAN Module

Thank you for choosing **RAK11310 WisBlock LoRaWAN Module** in your awesome IoT Project!

🎉 To help you get started, we have provided you with all the necessary documentation for your product.

- [RAK11310 Quick Start Guide](#) ↗
- [Datasheet](#) ↗
- [WisBlock Quick Start Guide](#) ↗
- [WisBlock Source Code Repository](#) ↗
- [Arduino IDE BSP](#) ↗
- [PlatformIO BSP](#) ↗
- [Bootloader source code and ready to flash file](#) ↗
- [RAK11310 3D Model](#) ↗
- [40-Pin Male Connector 3D File](#) ↗

Product Description

RAK11310 is a **WisBlock Core** module for RAK hardware IoT platform **WisBlock**. It is powered by an RP2040 ARM microcontroller developed by the Raspberry Pi Foundation, combined with the SX1262 LoRa transceiver from Semtech. **RAK11310** can be programmed via the Arduino™ IDE, PlatformIO, or MicroPython.

The RAK11310 WisBlock Core module has a RAK11300 LoRa stamp module in it together with a high-quality connector that is compatible with WisBlock Base boards. It allows an easy way to access the pins of the RAK11310 module in order to simplify the development of IoT devices.

The module complies with LoRaWAN 1.0.2 protocols, and also supports LoRa point-to-point communication.

The RF communication characteristic of the Lora module makes it suitable for a variety of applications in the IoT field such as home automation, sensor networks, building automation, personal area networks applications (health/fitness sensors, and monitors, etc.).

Product Features

- Based on RAK11300
- Uses the RP2040 as the main processor
- Semtech SX1262 low power high range LoRa transceiver
- LoRaWan 1.0.2 protocol stack
- **I/O ports:** UART/I2C/GPIO/USB
- Serial Wire Debug (SWD) interface
- **Module Size:** 20 × 30 mm
- **Supply Voltage:** 2.0 V ~ 3.6 V
- **Temperature Range:** -20 °C ~ 70 °C
- **Chipset:** Raspberry Pi Foundation RP2040, Semtech SX1262

Prerequisites

To use a **RAK11310**, you need at least a **WisBlock Base** to plug the module in. **WisBlock Base** is the power supply for the **RAK11310** module and has the programming/debug interface.

WARNING

- Make sure to fix the module with the screws to ensure a proper function.
- When using the LoRa module, make sure that an antenna is always connected. Using these transceivers without an antenna can damage the system.

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